process), and those jets shoot out through the galaxy and into intergalactic space, where they feed energy into the galaxy's huge radio-emitting lobes.

These black-hole-based explanations for quasars and radio galaxies are so successful that it is tempting to assert they *must* be right, and a galaxy's jets must be a unique signature crying out to us "I come from a black hole!" However, astrophysicists are a bit cautious. They would like a more ironclad case. It is still possible to explain all the observed properties of radio galaxies and quasars using an alternative, non–black-hole engine: a rapidly spinning, magnetized, supermassive star, one weighing millions or billions of times as much as the Sun—a type of star that has never been seen by astronomers, but that theory suggests might form at the centers of galaxies. Such a supermassive star would behave much like a hole's accretion disk. By contracting to a small size (but a size still larger than its critical circumference), it could release a huge amount of gravitational energy; that energy, by way of friction, could heat the star so it shines brightly like an accretion disk; and magnetic field lines anchored in the star could spin and fling out plasma in jets.

It might be that some radio galaxies or quasars are powered by such supermassive stars. However, the laws of physics insist that such a star should gradually contract to a smaller and smaller size, and then, as it nears its critical circumference, should implode to form a black hole. The star's total lifetime before implosion should be much less than the age of the Universe. This suggests that, although the youngest of radio galaxies and quasars *might* be powered by supermassive stars, older ones are almost certainly powered, instead, by gigantic *holes—almost* certainly, but not *absolutely* certainly. These arguments are not ironclad.

How common are gigantic black holes? Evidence, gradually accumulated during the 1980s, suggests that such holes inhabit not only the cores of most quasars and radio galaxies, but also the cores of most large, normal (non-radio) galaxies such as the Milky Way and Andromeda, and even the cores of some small galaxies such as Andromeda's dwarf companion, M32. In normal galaxies (the Milky Way, Andromeda, M32) the black hole presumably is surrounded by no accretion disk at all, or by only a tenuous disk that pours out only modest amounts of energy.